

AML Transaction Risk API

Presentation handout | What the system does and how one request moves through it

Assess one financial transaction for potential money-laundering risk, then return a review-ready decision with evidence.

Input

- Transaction amount, currencies, locations, payment type, timestamp
- Optional sender and receiver account history
- Validated by Pydantic before inference

Output

- Risk score from 0 to 100 and LOW to CRITICAL category
- Flag decision, confidence, threshold, warnings
- SHAP, LIME, and a controlled explanation/recommendation

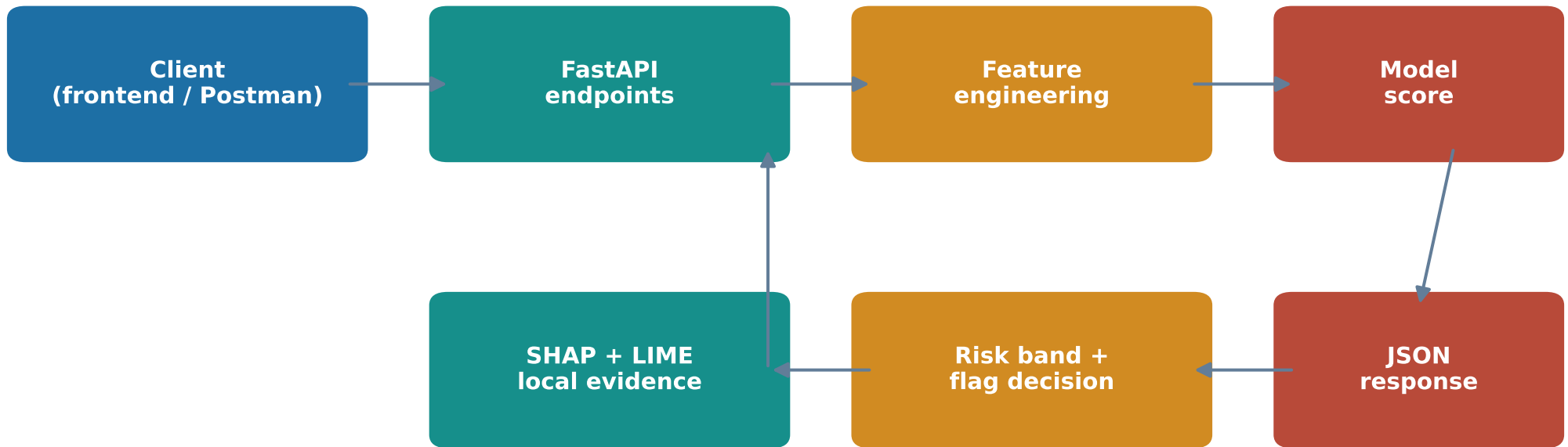
The central design choice

The model is recall-biased: missing a risky transaction is considered more costly than sending a false positive to human

Winning deployed method: Boosting only | LightGBM_scale_pos_weight | threshold 0.9574

System Architecture

The FastAPI module is the orchestration layer between clients, model artifacts, explainers, and optional Gemini text generation



Startup dependencies

lifespan() loads output/*.json, models/*, and splits/train.parquet. The model state is kept in the global ARTIFACTS object. If loading fails, /health stays available but scoring returns HTTP 503.

API Request Flow

One transaction from HTTP request to a compliance-oriented response

- 1 Validate**
Pydantic checks required fields, positive amount, category lengths, and timezone-aware timestamp.
- 2 Normalize**
Category values are uppercased; aliases map API values to the labels used during training.
- 3 Build 27 features**
Time, amount, cross-border, currency mismatch, encoded categories, account history, and ratios.
- 4 Score**
The selected ensemble returns a raw positive-class output. Current artifact configuration uses LightGBM boosting only.
- 5 Decide**
Raw output is compared with 0.9574; it is also mapped to a 0-100 operational score and risk band.
- 6 Explain**
SHAP explains the boosting arm; LIME approximates the final ensemble locally; Gemini is optional.
- 7 Respond**
RiskResponse returns score, flag, evidence, recommendation, warnings, IDs, and processing time.

CPU-bound work is sent through `asyncio.to_thread()`, so SHAP/LIME/model inference does not block the FastAPI event loop.

Features and Model Decision

The training contract is preserved at inference time

Feature groups

- Amount: raw amount, log amount, round-number indicators
- Time: hour, weekday, weekend, night, day of month
- Transaction: cross-border and currency mismatch
- Encoded categories: currencies, bank locations, payment type
- History: sender/receiver aggregates and amount ratios

Imbalance strategy

- No SMOTE or synthetic oversampling
- Boosting uses scale_pos_weight = 983.44
- Threshold selected from validation PR curve
- Chosen objective: max recall at precision ≥ 0.5
- Flag when raw output ≥ 0.9574

Risk score mapping

The raw model output is not presented as a calibrated probability. It is clipped to $[0, 1]$, compared with the validated flag threshold, and mapped into operational bands: below 50 LOW, 50-70 MEDIUM, 70-85 HIGH, and 85-100 CRITICAL. This keeps the displayed score, category, and binary flag aligned.

Explainability and Reliability

The system separates model evidence from generated language

SHAP

TreeExplainer runs against the winning boosting model. The response returns up to five strongest factors with feature value, impact, and direction. Positive impact means increases_risk; negative impact means decreases_risk.

LIME

A TabularExplainer is initialized from up to 2,000 sampled training rows. It perturbs the current vector and calls the full ensemble, so it describes the local final prediction.

Gemini

Optional. If GEMINI_API_KEY is configured and use_gemini is true, Gemini produces one concise explanation sentence. The recommendation remains policy-controlled by the API.

Fallbacks

Without Gemini, the deterministic formatter is used. SHAP/LIME failures become warnings and do not suppress a valid score. Missing artifacts produce degraded health and HTTP 503 for scoring.

Important framing: a risk flag is a review signal, not a finding that laundering occurred.

Endpoints and Example Contract

The service exposes health, a canonical assessment route, and two convenience/compatibility routes

Routes

- GET /health -> model status, artifact paths, Gemini status, loaded ensemble, timestamp
- POST /api/v1/risk-assessments/transactions -> canonical transaction assessment
- POST /predict -> alias for the canonical assessment route
- POST /api/v1/risk-assessments/transactions/{transaction_id} -> path ID overrides body ID
- Interactive documentation is available through FastAPI at /docs when the server is running

Request essentials

transaction_id (optional), sender_account, receiver_account, amount, payment_currency, received_currency, sender_bank_location, receiver_bank_location, payment_type, transaction_datetime, optional histories, use_gemini.

Response essentials

request_id, transaction_id, risk_score, raw_model_output, risk_category, model_confidence, should_flag, decision_threshold, ensemble_method, explanation, recommendation, warnings, processing_time_ms.

Run from ml_req: `uvicorn fastapi_app:app --host localhost --port 8500 --reload --env-file .env`

Artifacts and Results to Present

What is loaded, why it matters, and the measured held-out performance

Runtime artifacts

- output/split_info.json: 27 feature names and label encodings
- output/metrics_report.json: ensemble, threshold, metrics
- output/boosting_arm_metrics.json: winner and boosting configuration
- models/boosting_arm_model.joblib: deployed LightGBM model
- splits/train.parquet: LIME background data

Dataset snapshot

- Train: 6,690,227 rows; 6,796 positive
- Validation: 1,429,816 rows; 1,436 positive
- Test: 1,384,809 rows; 1,641 positive
- Highly imbalanced target, handled with weighting

Held-out test results

PR-AUC 0.7595 | precision at chosen threshold 0.7655 | recall 0.7599 | F1 0.7627. The chosen validation operating point was threshold 0.9574 with precision 0.6349 and recall 0.7521. The test result is a single held-out test-set touch, as noted in the metrics report.

Top global SHAP signals in the report: receiver_unique_senders, sender_min_amount, sender_unique_receivers, receiver_tx_count, sender_tx_count.

A Simple Explanation to Say Out Loud

Suggested 60-second project summary

This project exposes an AML transaction-risk model through a FastAPI service. A client sends transaction details and optional account history. The API validates and normalizes the input, reconstructs the same 27 features used in training, and sends them to the selected LightGBM model. Because laundering examples are rare, the model uses class weighting and a validation-selected threshold of 0.9574 rather than the default 0.5. The raw output is converted into an operational score and a LOW, MEDIUM, HIGH, or CRITICAL category, with a flag for analyst review. SHAP shows the strongest feature contributions, LIME explains the local ensemble behavior, and Gemini can turn that evidence into a concise sentence. The response is designed to support compliance review: it explains why a transaction was prioritized, but it does not claim that laundering occurred.

- Three points worth emphasizing**
- The API and model use the same feature/encoding contract.
 - Threshold selection reflects the business cost of missed risk.
 - Every assessment returns both a decision and supporting evidence.